

Sheth L.U.J.And Sir M.V. College of Science

Aim:-Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

Code:-

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*Practical4DS.py - C:/Users/info/OneDrive/Desktop/Practical4DS.py (3.14.6)
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import tkinter as tk
from tkinter import messagebox
import hashlib

def generate_keys():
    p = 61
    q = 53
    n = p * q
    phi = (p - 1) * (q - 1)
    e = 17
    d = pow(e, -1, phi)
    return (e, n), (d, n)

def create_signature(message, private_key):
    d, n = private_key
    message_hash = hashlib.sha256(message.encode()).hexdigest()
    hash_number = int(message_hash, 16)
    signature = pow(hash_number, d, n)
    return signature

def verify_signature(message, signature, public_key):
    e, n = public_key
    message_hash = hashlib.sha256(message.encode()).hexdigest()
    hash_number = int(message_hash, 16)
    recovered_hash = pow(signature, e, n)
    return recovered_hash == (hash_number % n)

def sign_message():
    message = message_entry.get("1.0", tk.END).strip()

    if not message:
        messagebox.showwarning("Warning", "Please enter a message.")
        return

    signature = create_signature(message, private_key)

    signature_entry.delete("1.0", tk.END)
    signature_entry.insert(tk.END, str(signature))

    result_label.config(
        text="Message signed successfully.",
        fg="green"
    )

def verify_message():
    message = message_entry.get("1.0", tk.END).strip()

    signature_text = signature_entry.get("1.0", tk.END).strip()

    if not message or not signature_text:
        messagebox.showwarning(
            "Warning",
            "Enter a message and generate a signature first."
        )
        return

    try:
        signature = int(signature_text)
    except ValueError:
        messagebox.showerror(
            "Error",
            "Invalid digital signature."
        )
        return

    if verify_signature(message, signature, public_key):
        result_label.config(
            text="✓ Signature Valid\nIntegrity and Authenticity Verified",
            fg="green"
        )
    else:
        result_label.config(
            text="✗ Signature Invalid\nMessage may have been modified",
            fg="red"
        )

def clear_all():
    message_entry.delete("1.0", tk.END)
    signature_entry.delete("1.0", tk.END)
    result_label.config(text="")

public_key, private_key = generate_keys()

root = tk.Tk()
root.title("RSA Digital Signature")
root.geometry("650x500")
root.resizable(False, False)

title = tk.Label(
    root,
    text="RSA Digital Signature",
    font=("Arial", 22, "bold")
)
```

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```
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File Edit Format Run Options Window Help
    font=("Arial", 22, "bold")
}
title.pack(pady=20)

message_label = tk.Label(
    root,
    text="Enter Message:",
    font=("Arial", 12, "bold")
)
message_label.pack()

message_entry = tk.Text(
    root,
    height=5,
    width=65,
    font=("Arial", 11)
)
message_entry.pack(pady=8)

signature_label = tk.Label(
    root,
    text="Digital Signature:",
    font=("Arial", 12, "bold")
)
signature_label.pack()

signature_entry = tk.Text(
    root,
    height=4,
    width=65,
    font=("Arial", 10)
)
signature_entry.pack(pady=8)

button_frame = tk.Frame(root)
button_frame.pack(pady=15)

sign_button = tk.Button(
    button_frame,
    text="Generate Signature",
    command=sign_message,
    font=("Arial", 11, "bold"),
    width=18
)
sign_button.grid(row=0, column=0, padx=8)

button_frame.pack(pady=15)

sign_button = tk.Button(
    button_frame,
    text="Generate Signature",
    command=sign_message,
    font=("Arial", 11, "bold"),
    width=18
)
sign_button.grid(row=0, column=0, padx=8)

verify_button = tk.Button(
    button_frame,
    text="Verify Signature",
    command=verify_message,
    font=("Arial", 11, "bold"),
    width=18
)
verify_button.grid(row=0, column=1, padx=8)

clear_button = tk.Button(
    button_frame,
    text="Clear",
    command=clear_all,
    font=("Arial", 11, "bold"),
    width=10
)
clear_button.grid(row=0, column=2, padx=8)

result_label = tk.Label(
    root,
    text="",
    font=("Arial", 13, "bold")
)
result_label.pack(pady=15)

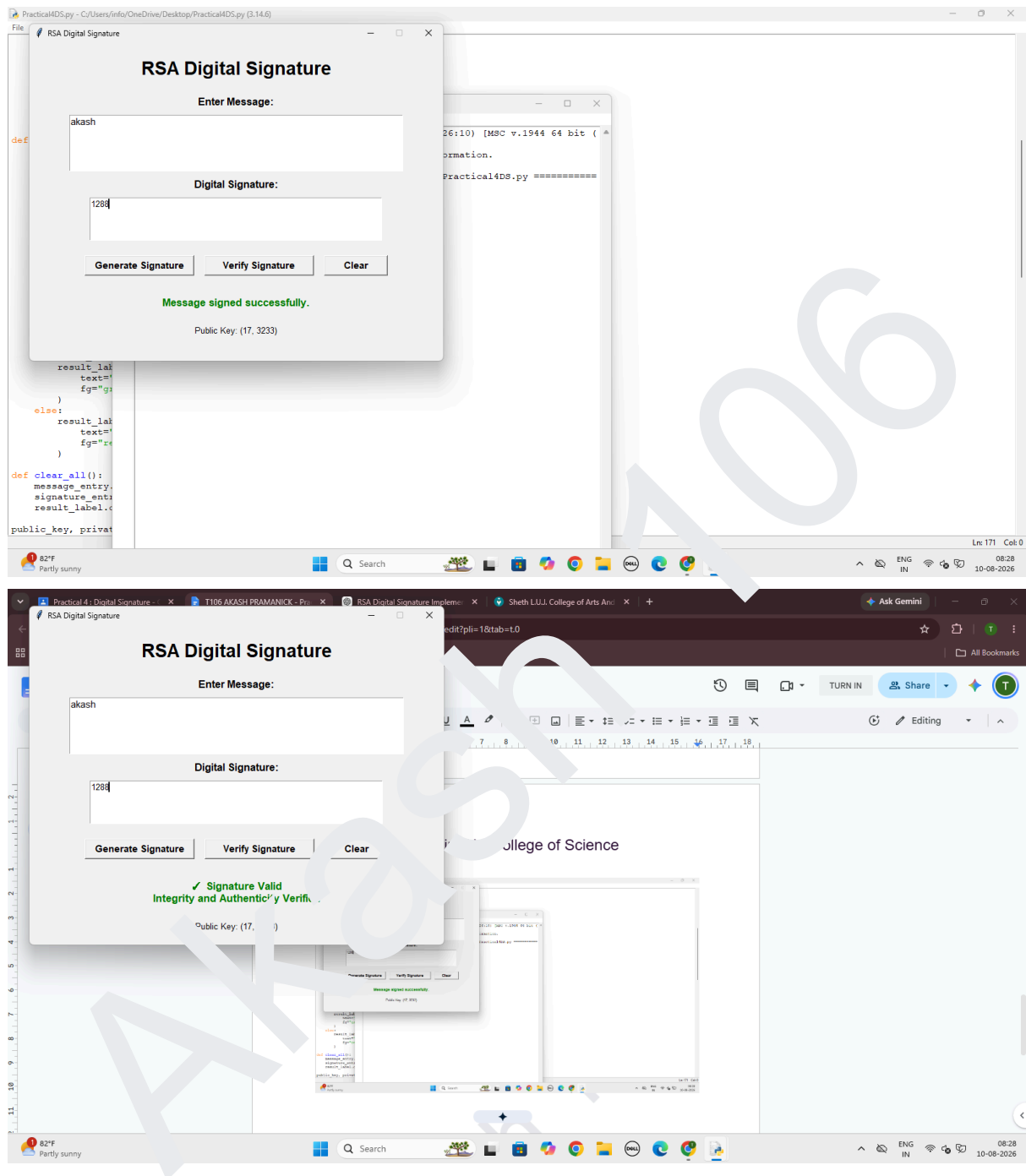
key_label = tk.Label(
    root,
    text=f"Public Key: {pubkey}",
    font=("Arial", 11)
)
key_label.pack(pady=15)

root.mainloop()
```

Output:-

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